

RecipeFusion: Recipe Recommendation System



**Tayyaba Ejaz
Laiba Hassan**

Supervised By
Dr. Sadia Ashraf

*Submitted for the partial fulfillment of BS Computer Science to the Faculty of
Engineering & Computer Science*

NATIONAL UNIVERSITY OF MODERN LANGUAGES

ISLAMABAD

May 2025

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CHAPTER 1

INTRODUCTION

1.1. Introduction:

Digital technologies have transformed how people approach everyday routines, including meal planning. However, many traditional recipe platforms offer limited personalization, lacking the ability to align suggestions with specific dietary needs, ingredient availability, or health goals. This results in a generic user experience that often fails to support those following unique dietary lifestyles—such as whole-food, plant-based diets—or individuals looking for sustainable and health-conscious options. These limitations create inefficiencies and extra effort for users who want tailored recipe recommendations that fit their preferences and resources.

“RecipeFusion: Intelligent Recipe Recommendation System” addresses this gap by offering a smart, user-friendly web platform that curates meal suggestions based on user input, including dietary restrictions, ingredient availability, and personal health goals. Leveraging artificial intelligence and data-driven analysis, the system provides personalized, nutritious recipes while offering in-depth nutritional information to aid informed decision-making. By making meal planning more intuitive and aligned with user lifestyles, RecipeFusion promotes healthier eating habits, reduces food waste, and supports a sustainable approach to nutrition.

1.2. Motivation:

The motivation behind RecipeFusion stems from the growing demand for intelligent, personalized, and health-focused digital meal planning solutions. In an age where health consciousness and sustainability are becoming central to daily life, traditional recipe platforms often fall short—especially for users with specific dietary needs or lifestyle preferences. These platforms typically offer one-size-fits-all content, making it difficult for individuals to discover meals that align with their nutritional goals, available ingredients, and unique dietary restrictions. As a result, users often face a frustrating and time-consuming process when trying to plan healthy meals that fit their routines.

According to a report by Statista, more than 70% of consumers express a desire to improve their diet, yet many struggle to find recipes that support these ambitions in a practical, personalized way. This gap presents a clear need for advanced tools that go beyond static search functions and provide dynamic, user-centric meal planning support.

Moreover, the global rise in health awareness, combined with a shift toward

sustainable living, has fueled interest in intelligent systems that encourage not only healthy eating habits but also environmentally responsible behavior. With the global health and wellness food market projected to exceed \$1 trillion by 2027, the potential to address this demand through smart, AI-powered digital tools is immense.

RecipeFusion aims to bridge this gap by offering an innovative solution that combines ingredient-based recipe discovery with comprehensive nutritional insights and support for diverse dietary preferences. By leveraging generative AI, it delivers meal recommendations that help users make informed, health-conscious decisions while simultaneously reducing food waste and promoting sustainable consumption. Through this approach, RecipeFusion not only enhances the user's cooking experience but also contributes to broader goals like public health improvement and environmental sustainability.

1.3. Problem Statement:

Traditional recipe discovery platforms fall short in addressing the growing demand for personalized, health-conscious, and efficient meal planning. Users often rely on generic search filters or keyword-based systems that do not account for individual dietary restrictions, available ingredients, or specific nutritional goals. This disconnect leads to frustration, as users struggle to find meals that truly align with their preferences and lifestyles.

According to a survey by the International Food Information Council, 63% of consumers actively seek healthy eating options, yet many are unsure how to identify meals that meet their personal needs. Furthermore, those following specialized diets—such as plant-based or allergen-free—face added difficulty navigating traditional platforms. The lack of integrated nutritional insights and customization results in time-consuming research, leading to poor planning, unhealthy choices, and increased food waste. These challenges highlight the need for a solution that offers intelligent recipe recommendations tailored to user input while supporting informed, health-driven, and sustainable eating habits.

1.3.1 Proposed Solution:

RecipeFusion offers a smart and personalized approach to recipe discovery by providing a platform that leverages artificial intelligence and user-specific dietary filters.

The system enables users to input available ingredients from their pantry or fridge. Select dietary preferences such as vegan, low-carb, or whole-food plant-based. Receive tailored recipe recommendations that align with their input. View detailed nutritional information for each recipe. Use an integrated meal planner to organize meals throughout the week. Interact with a conversational chatbot for recipe suggestions and guidance.

By combining convenience, personalization, and nutritional awareness, RecipeFusion aims to simplify the meal planning process and support users in making informed, health-focused, and sustainable food choices.

1.3.2 Objectives:

The main objectives of the RecipeFusion project are:

- To develop a recipe recommendation system that delivers personalized meal suggestions based on user preferences and available ingredients.

- To integrate AI-driven filtering for dietary needs such as vegan, low-carb, and whole-food plant-based diets.

- To provide comprehensive nutritional information for informed decision-making.

- To implement a user-friendly meal planner for organizing weekly meals.

- To include a conversational chatbot for enhanced user interaction and guidance.

- To promote sustainable eating habits by reducing food waste and encouraging mindful meal planning.

1.3.3 Scope of the Project:

The project focuses on the development of a web-based platform designed to enhance the meal planning experience through personalized recipe recommendations. Users can receive tailored meal suggestions based on their dietary preferences and available ingredients, with the support of intelligent AI-driven features. The system will offer a seamless and interactive experience, integrating tools that simplify meal preparation and promote healthy eating habits.

1.4. Goals and Objectives:

The goal of RecipeFusion is to develop an intelligent, web-based recipe recommendation system that simplifies meal planning through personalized suggestions based on user preferences, dietary needs, and available ingredients. The platform aims to

create a convenient, health-conscious, and engaging experience that encourages users to make informed and sustainable food choices.

The specific objectives are:

- To enable users to receive recipe suggestions based on ingredients they already have.

- To provide dietary filtering options for various preferences such as vegan, gluten-free, and whole-food plant-based diets.

- To incorporate nutritional information and support meal planning with a built-in grocery list generator.

- To include an AI-powered chatbot for interactive guidance and recipe discovery.

- To offer user account features such as registration, profile management, and personalized history.

- To promote sustainable food habits by reducing food waste through smart recipe matching.

- To enhance user satisfaction with a clean, intuitive, and user-friendly interface.

1.5. Scope of the study:

The scope of RecipeFusion encompasses the development of a web-based platform that delivers intelligent, personalized recipe recommendations and supports efficient meal planning.

The platform will provide users with the ability to.

- Search for recipes based on available ingredients using an intuitive interface.

- Filter recipes according to dietary preferences such as vegan, low-carb, and gluten-free.

- Generate custom recipes with the help of AI, based on user-input ingredients.

- Access nutritional information and use built-in meal planning tools with grocery list generation.

- Interact with a conversational chatbot for assistance in discovering recipes and planning meals.

- Register and manage user accounts for a personalized experience with saved preferences and history.

- Rate and review recipes to enhance future recommendations and system learning

1.6. Process model:

The Agile process model is chosen for the development of RecipeFusion due to its iterative, flexible, and user-centric approach. RecipeFusion is designed to be an intelligent and personalized recipe recommendation system that adapts to individual dietary needs, available ingredients, and health goals. These evolving and diverse user requirements necessitate a development methodology that supports continuous improvement and responsiveness to feedback—making Agile the ideal choice.

Agile enables rapid development and incremental delivery of key features, such as ingredient-based search, personalized recommendations, and nutritional analysis. This iterative nature allows the development team to build, test, and refine functionalities in short, manageable sprints. As new insights are gathered through user testing and feedback, Agile's flexibility ensures that enhancements or changes can be implemented quickly without disrupting the overall workflow.

Moreover, Agile fosters strong collaboration among cross-functional teams including UI/UX designers, front-end and back-end developers, and QA testers. This collaborative environment is essential for ensuring a smooth user experience and the technical reliability of RecipeFusion. Frequent sprint reviews and retrospectives help ensure that the system evolves in alignment with user expectations and technological best practices.

In essence, Agile supports the dynamic and evolving nature of RecipeFusion, allowing it to grow into a powerful, user-centered tool that promotes healthy, sustainable eating while continuously improving based on real-world user interactions.

1.7. Nature of the project:

The nature of the project is primarily web-based and focuses on personalized recipe recommendations and meal planning within the health and wellness domain. The project integrates artificial intelligence, user preference filters, and intuitive UI components to deliver an engaging and intelligent platform for modern, health-conscious users. Key features of the platform include

Smart recipe discovery based on user-inputted ingredients, dietary restrictions, and health objectives. Interactive recipe pages that provide detailed cooking instructions, nutritional breakdowns, and tips for healthy and sustainable meal preparation. User

engagement features, such as saving favorite recipes, submitting reviews, and tracking personal searches to enhance future recommendations.

The backend is developed using Node.js and MongoDB, ensuring a robust and scalable foundation for handling user data, recipes, reviews, and personalized preferences. Security and performance are prioritized through the use of JWT-based authentication, role-based access control, and optimized API design.

1.8. Overview/Organization of the Project:

The report is structured as follows:

Chapter 1: Provides an introduction to the project, detailing its motivation, problem statement, goals, objectives, scope, and nature of the project.

Chapter 2: Discusses relevant background knowledge, reviews existing systems and technologies, and analyzes their limitations in relation to the proposed solution.

Chapter 3: Specifies the functional and non-functional requirements of the system, outlining what the system should achieve and the technical standards it must meet.

Chapter 4: Focuses on the system design, including architectural views, database modeling, and user interface designs.

Chapter 5: Describes the implementation process, detailing algorithms, modules, and technologies used in the development of the platform.

Chapter 6: Presents results, testing procedures, and validation of the system, along with performance analysis.

Chapter 7: Concludes the report, summarizing the project's achievements, identifying limitations, and providing recommendations for future improvements.

CHAPTER 2

EXISTING SYSTEM

2.1 Introduction:

This chapter explores the background and existing systems related to recipe recommendation and meal planning platforms. It provides insight into important concepts, technologies, and limitations of current solutions, establishing the relevance and innovation of RecipeFusion in addressing these gaps.

2.2 Explanation of important constructs of the Application Domain:

2.2.1 Personalized Recipe Recommendations:

Personalized recipe platforms aim to offer meal suggestions tailored to individual users based on their dietary preferences, health goals, and available ingredients. The systems enhance user experience by promoting relevant, time-saving options and reducing the need for manual search or filtering.

2.2.2 Ingredient-Based Search:

At the heart of the application is a system that allows users to input available ingredients and receive tailored recipe suggestions. This construct supports the idea of minimizing food waste and simplifying the cooking process by making use of what users already have on hand. It also enhances convenience by eliminating the need to shop for additional ingredients unnecessarily.

2.2.3 User Interaction and Feedback:

Users can save favorite recipes, submit reviews, and track previous searches. This construct not only improves user engagement and retention but also enables the system to refine future recommendations through feedback loops and usage patterns.

2.2.4 Dietary Preference Filters:

To accommodate a variety of eating lifestyles and restrictions, the system includes filtering mechanisms that allow users to narrow down recipe results based on dietary needs. This supports inclusivity and helps ensure that users receive relevant results tailored to their specific requirements.

2.2.5 Key features of proposed system:

The proposed platform will offer the following essential features:

- Ingredient based search and smart suggestions
- Filtering options for vegan, vegetarians, gluten-free and other
- Custom recipes based on user input
- User account management and recipe history tracking

2.3 Existing Studies/Systems:

Several existing systems offer recipe discovery and meal planning functionalities, each with its own strengths and limitations. While they demonstrate the usefulness of digital tools in enhancing dietary decision-making, most fall short in delivering a truly intelligent and personalized experience. The table below highlights some well-known systems, their features, technologies, and identified gaps that RecipeFusion aims to address.

Year	System	Contribution	Technologies Used	Limitations
2014	Yummly [1]	Recipe recommendations based on preferences	Web, Mobile App, Basic AI	Limited personalization, lacks ingredient-based search
2012	AllRecipes [2]	Ingredient search and recipe ratings	Web Technologies	No AI-generated recipes, no meal planner
2013	Eat This Much [3]	Calorie-based meal planning	Web, Nutrition API	Focuses on calories, not full dietary needs
2021	ChefGPT [4] [5]	AI-generated recipes using user input	AI/ML, NLP	Lacks integrated meal planning or nutrition tracking

These systems highlight the increasing interest in intelligent meal planning tools. However, they often lack holistic features such as ingredient-aware suggestions, integrated nutrition analysis, dynamic AI-driven recipe generation, and personalized chat-based interaction. RecipeFusion aims to fill these gaps by offering a comprehensive, user-centric platform.

2.4 Comparison of Existing Systems:

Feature	Yummly	All Recipes	Eat this much	Chef GPT	Proposed System
AI Personalized	Basic	None	Low	Yes	Yes
Ingredient Based Search	No	Yes	No	Partial	Yes
Nutritional Information	Basic	Limited	Yes	No	Yes
Meal Planner	No	No	Yes	No	Yes
Chatbot support	No	No	No	Partial	Yes
Custom recipe generation	No	No	No	Yes	Yes
User Experience	Good	Medium	Medium	Good	High

2.5 Summary:

This chapter reviewed the domain knowledge and existing systems relevant to the project. While several platforms offer recipe recommendations and meal planning features, few deliver a fully personalized, ingredient-aware, and nutrition-integrated experience. The proposed system RecipeFusion leverages modern technologies such as artificial intelligence, real-time ingredient-based filtering, and interactive chatbot support to fill these gaps. By combining personalization, usability, and health-conscious features, RecipeFusion provides a more intelligent, scalable, and user-friendly solution for modern dietary needs

CHAPTER 3

REQUIREMENTS SPECIFICATION

3.1 Introduction:

This chapter outlines the requirements necessary to develop the **RecipeFusion recipe recommendation system**. These requirements are categorized into interface requirements—ensuring seamless communication between different system components—and functional requirements, which define the core features the system must deliver. Additionally, the chapter distinguishes between hardware and software interface requirements to provide a comprehensive understanding of the system’s operational and technical needs. Together, these requirements ensure that RecipeFusion remains efficient, scalable, and aligned with user expectations for intelligent, personalized meal planning.

3.2 Interface Requirements:

Interface requirements specify the essential elements needed to ensure smooth communication between various components of the system. These requirements define how different modules—such as the recommendation engine, user interface, and external APIs—interact and integrate effectively. Additionally, this section outlines how the system interfaces with external environments, including web browsers, cloud services, and third-party data sources.

3.2.1 Software Interface Requirements:

To ensure smooth operation and accessibility of the system, the following resources are needed:

User-End Devices:

Users accessing the recipe application, especially for rich media interaction (e.g., uploading images, viewing charts, filtering recipes), should use devices that meet the following minimum requirements:

Processor: Minimum quad-core, 2.5 GHz or higher

RAM: At least 16 GB to support smooth rendering

Storage: SSD with a minimum of 256 GB of free space for fast loading

Cloud or Local Server:

The backend MERN-based system requires a stable and scalable server infrastructure to support user authentication, real-time data fetching, and administrative analytics.

Processor: Multi-Core (Quad-Core or better) to handle concurrent APIs

requests

RAM: 16GB or more to manage multiple user sessions concurrently, database queries, and file operations

Storage: At least 1 TB SSD storage to store recipes, reviews, user information and usage analytics

Connectivity: High-speed internet connection to ensure fast and consistent API responses.

3.2.2 Hardware Interface Requirements:

Frontend Technologies:

ReactJs: For building a responsive and dynamic UI.

Backend Technologies:

NodeJs: Handles API logic, authentication and other tasks.

ExpressJs: Provides routing and middleware support for API development.

MongoDB: A NoSQL database used for storing user and roles, recipe and ingredients, reviews and ratings, and search history details.

External APIs:

Email/OTP: Using nodemailer for secure OTP delivery and password reset flow.

Operating System Capabilities:

Designed for cross-platform support, allowing access from:

Desktops and Laptops: Windows 10/11, macOS, Ubuntu/Linux

Mobile Devices: Android and iOS (via responsive web UI)

3.3 Functional Requirements:

Interface requirements specify the essential elements needed to ensure smooth communication between various components of the system

3.3.2 Authentication and Authorization

Users must be able to register, log in, and log out securely. Access to saving recipes, receiving suggestions and making searches need authenticated access.

3.3.3 Recipe Discovery

Enable filtering by dietary preferences (e.g., vegan, vegetarian, gluten-free, etc).

3.3.4 Custom Recipe Generation

Generate custom recipes by searching with the available ingredients and preferences that align with health goals

3.3.5 Shopping List Generator

Automatically generate grocery list based on the searched recipe

3.3.6 Recipe Saving and retrieval

Allows user to save the favorite recipes and enable easy access for further use or retrieval

3.3.7 Responsive Design

Ensures compatibility with desktops, laptops, tablets and mobile devices. Deliver a seamless experience regardless of screen size or device.

3.3.8 Performance and Scalability

Handle multiple concurrent users without performance degradation. Support further expansion with scalable architecture.

By fulfilling these requirements, RecipeFusion will deliver a robust, intelligent platform that simplifies meal planning, encourages healthy choices, and enhances user satisfaction through personalization and interactivity.

3.4 Use Case Model:

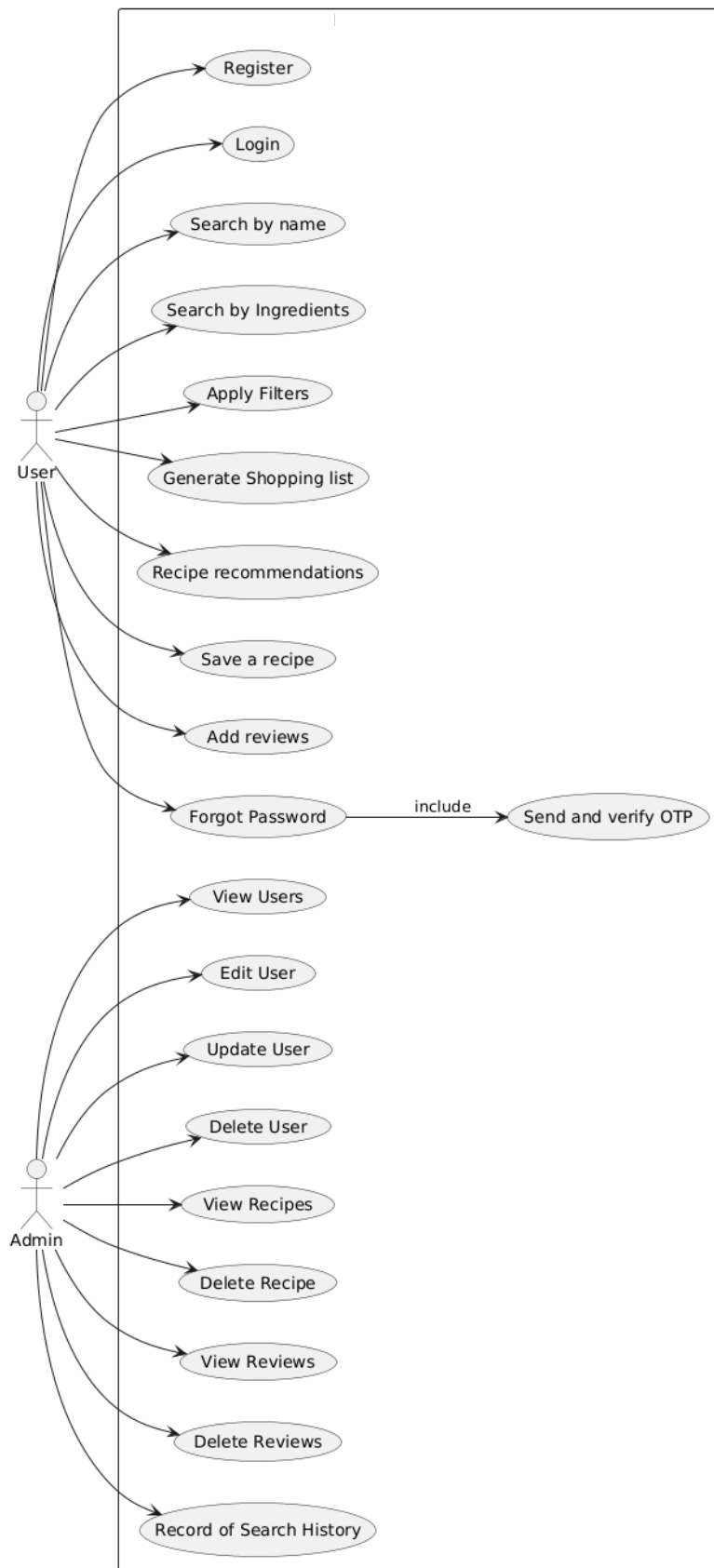


Fig 3.4 – Use Case Diagram

3.5 Use Cases:

3.5.1 Use Case 1: Register

The register use case enable new users to create an account on the platform. Registration is mandatory to access several features. For registration user has to register by providing basic credentials like username, email, phone, gender and password and enable user to login for accessing the resources

Aspect	Details
Actor	User
Description	Allows user to register by entering details such as username, email, phone, gender and password.
Precondition	User is not logged in and complete features are not accessible to that user.
Trigger	User clicks the “Sign up” button to get registered.
Postcondition	Account is already created and user is registered just need to login instead of register.
Primary Flow	- User fills the signup form with complete details. - The system validates data given by user. - System stores the user’s information in database. - User receives confirmation message.
Alternate Flow	If username or email already exists error message will be shown to user
Includes/Extends	Input validation

3.5.2 Use Case 2: Login

This use case enables registered users to authenticate themselves and gain access to personalized features of the application. A successful login session allows the user to search, save recipes, and more.

Aspect	Details
Actor	User
Description	Allows registered users to log in using their email and password
Precondition	User is registered for the account.

Trigger	User clicks the “Login” button and submits credentials.
Postcondition	User is logged in and redirected to the respective dashboard
Primary Flow	- User enters email and password. - The system verifies the credentials. - If verification is successful user is redirected to respective dashboard e.g; user or admin based on role (1 for admin and 2 for user).
Alternate Flow	If credentials are incorrect systems shows the error message.
Includes/Extends	Authentication logic

3.5.3 Use Case 3: Search by Name

This use case allows users to find specific recipes by entering a recipe name. It is especially useful when the user knows what they are looking for.

Aspect	Details
Actor	User
Description	Enables the user to find recipes based on a specific recipe name.
Precondition	Must be logged in.
Trigger	User enters name in search bar (can select filters too) and clicks “Search” button.
Postcondition	Matching recipes are displayed on screen
Primary Flow	- User type recipe name. - The system searches for the matching recipe name. - Displays the results
Alternate Flow	No recipe found → system shows "No recipes found".
Includes/Extends	Text-search

3.5.4 Use Case 4: Search by Ingredients

Allows users to discover recipes using available ingredients, helping reduce food waste and enhance convenience.

Aspect	Details
Actor	User

Description	Enable users search for recipes based on ingredients they have.
Precondition	Must be logged in.
Trigger	User types ingredients and clicks “Search” button.
Postcondition	List of recipes using those ingredients is displayed.
Primary Flow	- User inputs ingredients. - The system matches recipes with those ingredients. - Displays the results
Alternate Flow	If no matches → system suggests related recipes or asks to broaden search.
Includes/Extends	Ingredient-matching logic

3.5.5 Use Case 5: Apply Filters

This use case enhances recipe searching by letting users apply dietary, preparation time, or cuisine-based filters.

Aspect	Details
Actor	User
Description	Allows users to refine recipe searches with dietary or cuisine filters.
Precondition	User is about to perform a search.
Trigger	User selects any filter options.
Postcondition	Search results are updated with selected filters.
Primary Flow	- User selects filters - The system search with that filter. - Filtered results shown.
Alternate Flow	No matches found → show "No results".
Includes/Extends	Filtering logic

3.5.6 Use Case 6: Generate Shopping List

This use case generates a grocery list from selected recipes, helping users plan and shop efficiently.

Aspect	Details
Actor	User
Description	Creates a shopping list from ingredients used in selected recipe.
Precondition	User has selected a recipes.
Trigger	User clicks “Generate Shopping List” button.
Postcondition	List of ingredients is displayed.
Primary Flow	- User views a recipe - The system provides ingredient list.
Alternate Flow	No recipes selected → prompt user to select at least one recipe
Includes/Extends	-

3.5.7 Use Case 7: Recipe Recommendations

This use case suggests recipes to users based on selected ingredients, filters, or previous behavior. It aims to provide users with convenient and personalized recipe suggestions.

Aspect	Details
Actor	User
Description	Recommends recipes based on user preferences.
Precondition	User is logged in.
Trigger	System automatically suggests.
Postcondition	Recommended recipes are displayed on screen
Primary Flow	- System fetches and displays personalized recommendations.
Alternate Flow	System suggests popular or trending recipes.
Includes/Extends	-

3.5.8 Use Case 8: Save a Recipe

Allows users to bookmark recipes for later use, making it easier to retrieve favorite or interesting recipes. The saved recipe will be saved in the user’s favorite list and he can retrieve whenever he wants

Aspect	Details
Actor	User
Description	Enables users to save recipes to their personal account..
Precondition	User is logged in and viewing a recipe.
Trigger	User clicks the “Save” or bookmark icon on a recipe.
Postcondition	Recipe is stored in user’s saved recipes list.
Primary Flow	- User clicks the save button - System stores recipe ID under user’s profile - User gets success message
Alternate Flow	System shows ”Already saved”
Includes/Extends	Save/Remove from saved logic

3.5.9 Use Case 8: Add review

Allows users to give review on a specific recipe for later use. User can express his thoughts whether he likes or dislikes the recipe.

Aspect	Details
Actor	User
Description	Enables users to give review on a recipe.
Precondition	User is logged in and viewing a recipe.
Trigger	User clicks the “Add review” button on the detail page of a recipe.
Postcondition	Review is saved and associated with the recipe.
Primary Flow	- User enters the review. - System stores the review - User gets success message
Alternate Flow	Incomplete form
Includes/Extends	Input validation.

3.5.10 Use Case 8: Admin Functions

View Users

Aspect	Details
Actor	Admin
Description	Displays all users registered on the platform for admin reference.
Precondition	Admin is logged in.
Trigger	Admin navigates to user management page.
Postcondition	User list is displayed.
Primary Flow	- Admin logs in - Admin opens “All Users” view - System retrieves and displays data
Alternate Flow	System/database error shows the appropriate message.
Includes/Extends	Loading users.

Edit/Update/Delete Users

Aspect	Details
Actor	Admin
Description	Allows admins to edit or remove user records.
Precondition	Admin is logged in and viewing users list.
Trigger	Admin clicks edit or delete next to a user.
Postcondition	User information is updated or removed.
Primary Flow	- Admin selects a user. - Admin clicks edit or delete. - System performs update/removal.
Alternate Flow	User doesn’t exit or system error.
Includes/Extends	Form validation, authorization check.

View/Delete Recipes

Aspect	Details
Actor	Admin

Description	Provides a list of all recipes and allows deletion if necessary.
Precondition	Admin is logged in.
Trigger	Admin accesses recipe management.
Postcondition	Recipes are listed or deleted after selection.
Primary Flow	- Admin opens recipe list. - Admin clicks delete any recipe. - System updates database.
Alternate Flow	Recipes not found.
Includes/Extends	Retrieving recipes, delete confirmation.

View/Delete Reviews

Aspect	Details
Actor	Admin
Description	Allows viewing and deleting user reviews.
Precondition	Admin is logged in.
Trigger	Admin accesses review management section.
Postcondition	Review is removed by admin if inappropriate.
Primary Flow	- Admin opens reviews list. - Identify any review to delete. - System remove that review.
Alternate Flow	Review doesn't exist or deletion fails.
Includes/Extends	Viewing and deletion logic.

View Search History

Aspect	Details
Actor	Admin
Description	Displays a log of past user searches to analyze patterns
Precondition	Admin is logged in.

Trigger	Admin accesses the analytics or history section.
Postcondition	System displays search trends or user-specific history.
Primary Flow	- Admin opens main tab. - System loads and displays history data.
Alternate Flow	Data unavailable → show fallback message.
Includes/Extends	Statistics loading

3.6 Non-Functional Requirements:

Specifies the quality attributes the system must meet to ensure proper operation beyond functional correctness. It focuses on the platform's responsiveness, user accessibility, reliability, and security when recommending recipes and tracing ingredient sources.

3.6.1 Performance:

The system must ensure that key features, such as search any recipe and get details of that recipe are rendered and accessible within 3 seconds from the time a user initiates a request.

The application should be optimized to maintain minimal load times, even under high traffic or during peak usage hours, to ensure consistent and efficient service delivery.

3.6.2 Reliability:

The platform must maintain 99.9% uptime, which equates to less than 1 hour of downtime per year. This requirement covers all application components, including the backend services and database.

Regular automatic backups and failover mechanisms should be in place to protect against data loss and allow rapid recovery in the event of server failures or system crashes.

3.6.3 Security:

All communication between client and server must be secured using HTTPS, ensuring that sensitive user data is encrypted and protected against interception or man-

in-the-middle attacks.

User passwords must be hashed and securely stored using strong hashing algorithms. Security features will include:

- Password reset functionality

- Session management

- Role-based access control

3.6.4 Consistency:

RecipeFusion must maintain data consistency across all user interactions. Any updates made to user preferences, saved recipes, or meal plans must be reflected in real-time across the system.

3.7 Resource Requirements:

The resource requirements define the essential technical and infrastructural components necessary to support the platform's ingredient-based recipe discovery, smart contract-based transactions, and media handling. It includes specifications for user-end devices, server environments, development tools, and external services.

3.7.1 Equipment:

Hardware: The system will require a high-performance server capable of handling multiple users interacting.

Software: The development environment will be built using contemporary web technologies. Visual Studio Code will serve as the main integrated development environment (IDE) for coding tasks. Node.js will be utilized as the server-side runtime environment, with Express.js managing the backend logic and routing. MongoDB, a flexible and scalable NoSQL database, will be employed to store user profiles, product information, and related data efficiently.

3.7.2 Human Effort:

Frontend Development (40% effort): This phase focuses on designing and implementing the user interface to ensure an intuitive, seamless experience. Developers will work extensively with React.js and APIs to build recipe

browsing, filtering and dashboards.

Backend Development (30% effort): Backend engineers will design and implement the server architecture, including APIs, user authentication, and database schema. The work involves optimizing data flow, ensuring security measures, and managing integration points for APIs. Leveraging the MERN stack, the backend will ensure efficient, scalable handling of requests and data persistence

Testing and Quality Assurance (20% effort): Testing will cover functional correctness, security vulnerabilities, and responsiveness across different devices and network conditions.

Deployment (10% effort): This phase involves configuring and launching the application to a live production environment. Includes setting up cloud infrastructure (such as vercel, render or AWS), managing environment variables. Developers will ensure a secure, smooth deployment process and monitor system performance.

3.8 Database Requirements:

3.8.1 MongoDB:

The database will include following collections:

Users: Stores information for registered users like username, email, phone number, gender, role, password, etc.

Reviews: Stores information for the reviews by users like userId, recipeId, title, review, sentiment, etc.

Recipes: Stores information for the recipes name, image, category, description, ingredients, steps, etc.

SavedRecipes: Stores information for the recipes saved by users like userId, recipeId, title, image, etc.

SearchHistory: Stores information for the search history done by users like userId, query, type etc.

3.9 Project Feasibility:

The study assesses whether the project is technically viable, economically justified, and operationally practical. It considers the resources, technologies, and potential constraints involved to determine if the system can be successfully implemented and sustained over time. The feasibility analysis is crucial to ensure that the system meets user needs while aligning with project goals and available resources.

3.9.1 Technical Feasibility:

The development of the **RecipeFusion** system is technically feasible due to the maturity and stability of modern web development technologies. The use of the **MERN stack** (MongoDB, Express.js, React.js, Node.js) provides a robust and scalable foundation for building both the backend and frontend of the application. React enables the development of an interactive and responsive user interface, while Node.js and Express.js ensure efficient handling API logic.

3.9.2 Operational Feasibility:

RecipeFusion is designed with a user-first approach that prioritizes performance and usability. Key workflows such as ingredient-based search, meal planning, and recipe generation are optimized to ensure fast response times. The intuitive layout, real-time feedback (e.g., during chatbot interaction), and minimal page load times contribute to a seamless user experience.

Operationally, the platform will require moderate server maintenance, routine database backups, and occasional feature updates. With a well-structured backend and a responsive frontend, the system is expected to perform reliably even under increasing user demand.

3.9.3 Legal and Ethical Feasibility:

RecipeFusion will strictly adhere to data protection laws and ethical guidelines related to user privacy and digital responsibility. Sensitive user data, such as email addresses and passwords, will be encrypted and stored securely using hashing algorithms like bcrypt.js. The system will implement access control mechanisms and secure sessions to prevent unauthorized access.

3.10 Summary:

This chapter outlined the essential non-functional requirements for the **RecipeFusion recipe recommendation platform**, focusing on aspects such as performance, scalability, security, and reliability. These elements are critical to ensuring that the system delivers a responsive, safe, and consistent experience for users while supporting multiple concurrent sessions and maintaining data accuracy.

Additionally, the chapter covered the required resources—both technical and human—necessary for the successful development and maintenance of the platform. It emphasized the use of modern, well-supported technologies such as React.js, Node.js, and MongoDB, which collectively provide a stable and scalable foundation for RecipeFusion.

Finally, the feasibility analysis confirmed that the project is technically, operationally, and ethically sound. With a user-centric design and a robust technical framework, RecipeFusion is well-positioned to meet its objectives and provide an intelligent, personalized solution for modern meal planning and healthy eating. These considerations collectively establish a strong base for the successful implementation of the system.

CHAPTER 4

SYSTEM MODELING

4.1 Introduction:

System modeling plays a pivotal role in software development, acting as a bridge between high-level conceptualization and real-world implementation. It enables a structured and visual representation of the system's architecture, workflows, data exchange, and user interactions, providing clarity and guidance for developers, designers, and stakeholders alike.

This chapter presents the system design of RecipeFusion — a web-based recipe recommendation platform that empowers users to discover meal options tailored to their dietary preferences, available ingredients, and past feedback. The platform aims to simplify the cooking experience, encourage healthy eating, and promote ingredient transparency by integrating smart recommendations and traceable data flows.

The primary goal of RecipeFusion is to offer users an intelligent, personalized, and intuitive interface to explore and save recipes while considering their nutritional goals, preferences (e.g., vegan, gluten-free), and ingredient availability. The system also supports basic traceability features by allowing users to view source details of key ingredients when available.

To ensure an effective, scalable, and user-centered architecture, several modeling techniques have been adopted. These include:

Class Diagram – Illustrates the core entities of the system such as Users, Recipes, Ingredients, and Reviews, along with their relationships.

Activity Diagram – Depicts the flow of user actions such as recipe searching, saving, reviewing, and filtering based on available ingredients.

System Sequence Diagram – Visualizes the sequence of interactions between users and the system during key operations like login, recommendation generation, and review submission.

Component Diagram – Shows the modular organization of the system, covering the frontend interface, backend APIs, database layer, and third-party services (e.g., Cloudinary, authentication).

Deployment Diagram – Represents the physical architecture, including client

devices, cloud-based frontend hosting (e.g., Vercel), backend deployment (e.g., Render or AWS), and database servers.

Data Flow Diagram (DFD) – Maps how data moves through the system, from user input to data retrieval, recommendation engine processing, and result display.

Together, these models provide a comprehensive view of the internal mechanisms and external interactions of the RecipeFusion system, ensuring a well-documented foundation for implementation, scalability, and future enhancements.

4.2 System Design:

Designing the system architecture is a foundational step in the development of the RecipeFusion platform. It involves defining the structural blueprint of the system and determining how different modules and services interact to achieve the overall goal of intelligent and personalized recipe recommendations.

The design process begins with user interaction through a responsive web interface. Users can perform a range of actions including signing up, searching with dietary preferences, searching for recipes, and providing feedback. These interactions are processed through an application backend that coordinates with a machine learning-based recommendation engine and a cloud-hosted database system.

User inputs such as ingredient availability, meal preferences, and dietary restrictions are collected and processed in real time. This data is then utilized by the recommendation engine, which generates recipe suggestions by applying collaborative filtering and content-based filtering techniques. The suggested recipes are retrieved from a structured recipe database and presented back to the user through the web interface.

The overall design of RecipeFusion emphasizes modularity, scalability, and user-centric personalization. The system is built to support continuous learning through user feedback, thereby refining the accuracy of future recommendations. In addition, security and data privacy are maintained through encrypted communication and secure data storage practices.

4.3 Design Approach:

For the development of the RecipeFusion platform, a bottom-up design approach was adopted, prioritizing the construction and validation of core functional modules before

assembling them into a cohesive application. The process began with the design and setup of the backend infrastructure using Node.js, Express.js, and MongoDB, which formed the foundation for user management, recipe data handling, and API endpoints. Parallel to this, the frontend interface was developed using React.js, focusing on creating an interactive and user-friendly environment for browsing, filtering, and saving recipes.

Each module—whether frontend or backend—was independently developed and thoroughly tested to ensure functionality and stability. Integration was performed incrementally, allowing seamless connection between the user interface, API services, and the underlying database. This structured approach not only enhanced reliability and maintainability but also simplified debugging and quality assurance efforts across the development lifecycle. By verifying each component in isolation before integration, the team ensured a smooth and consistent experience for end users while enabling future scalability and feature expansion.

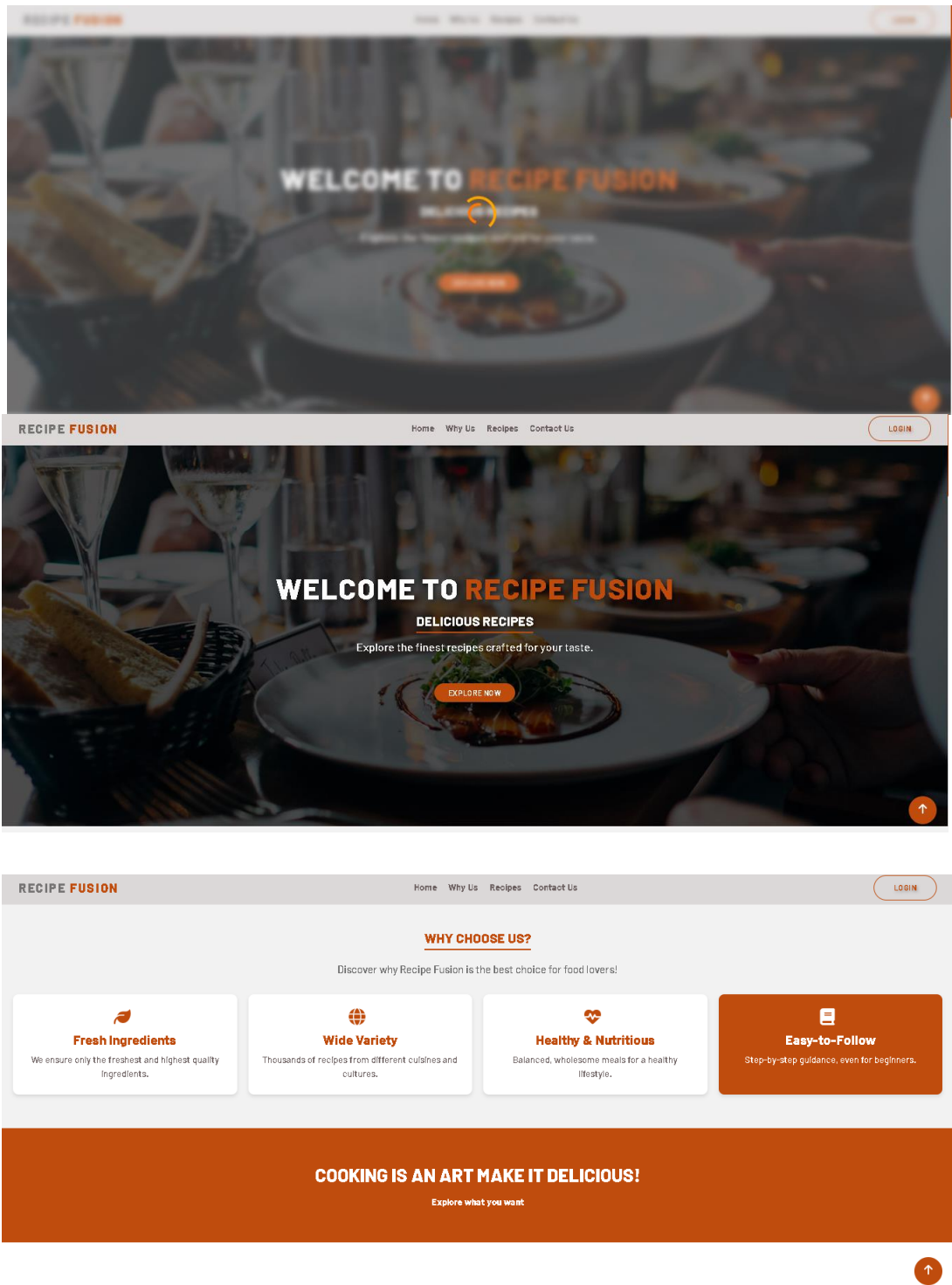
4.4 Interface Design:

The user interface design of the RecipeFusion system focuses on clarity, ease of use, and accessibility for all user types, including casual users, nutrition-focused individuals, and administrators. A clean layout and intuitive navigation structure were adopted to ensure users can interact with the system efficiently. The interface includes components for signing up, logging in, setting dietary preferences, browsing recipes, and providing feedback. Throughout the design process, mockups and prototypes were shared with users to gather feedback and refine usability. The final interface aligns with UI/UX best practices, ensuring both functionality and visual appeal.

4.4.1. High-Fidelity Prototype:

Below are the screenshots representing the UI of the platform:

Landing Screen:



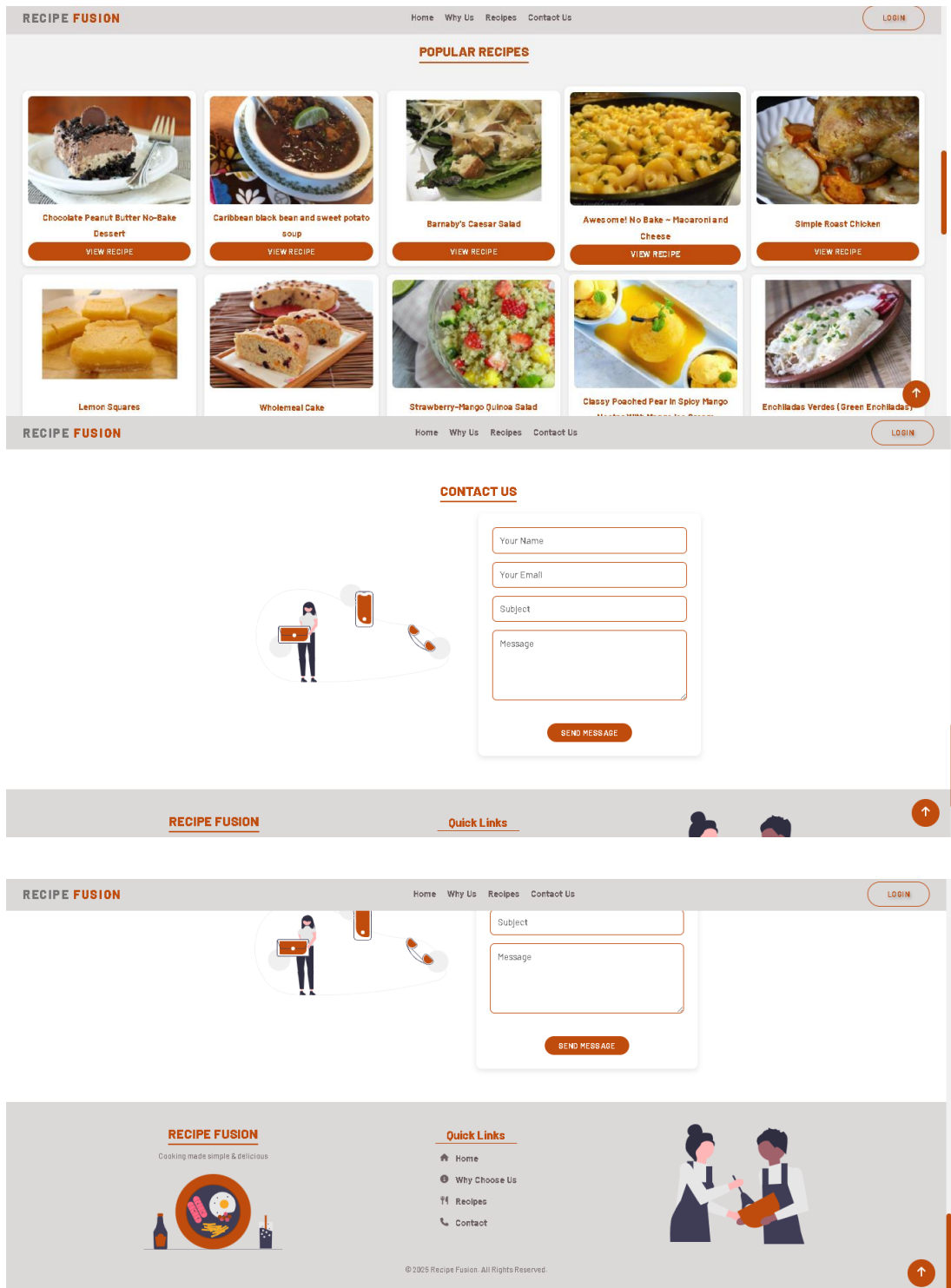


Fig 4.4.1 (a) – Landing Page

Login Screen: Users login from here to access their respective dashboards

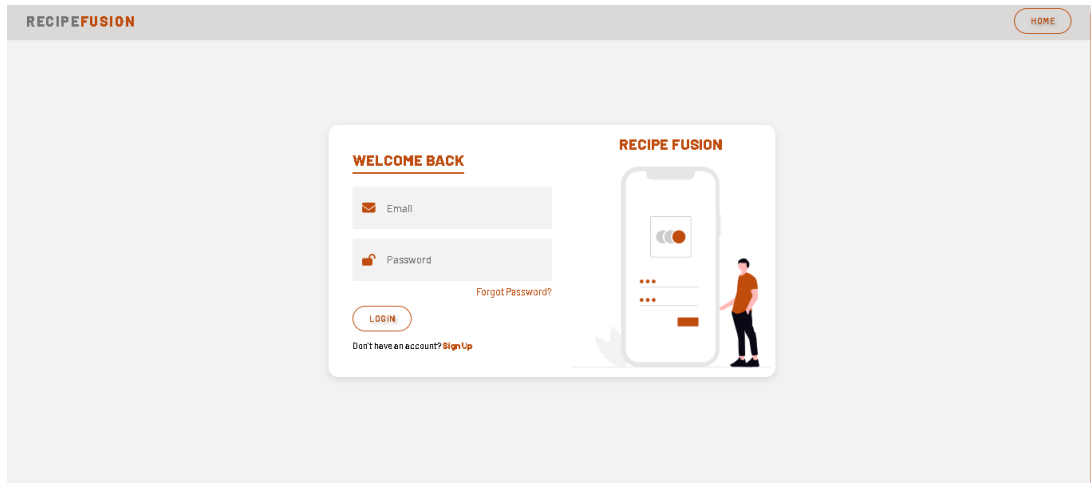


Fig 4.4.1 (b) – Login Screen

Signup Screen: Users can register themselves to access particular features of the system

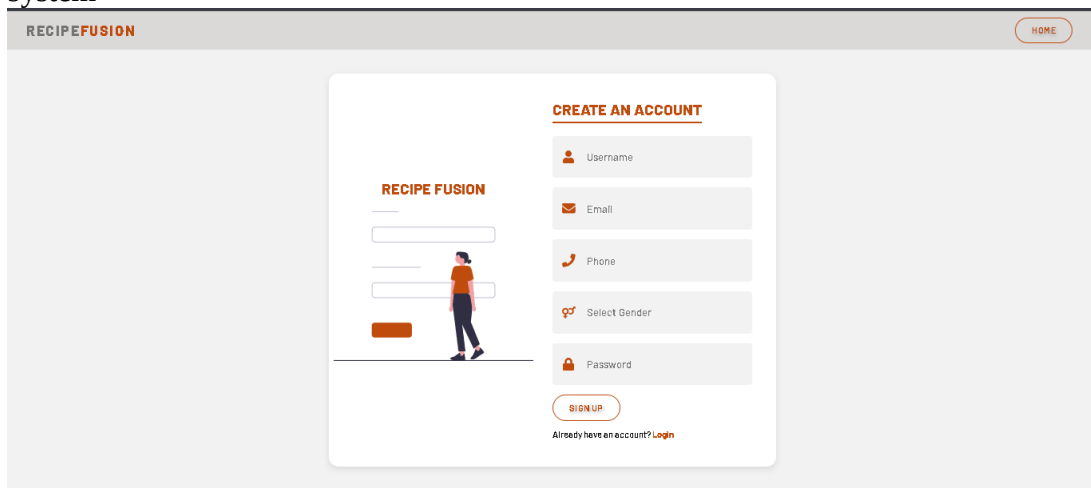


Fig 4.4.1 (c) – Signup Screen

Admin Dashboard Screens:

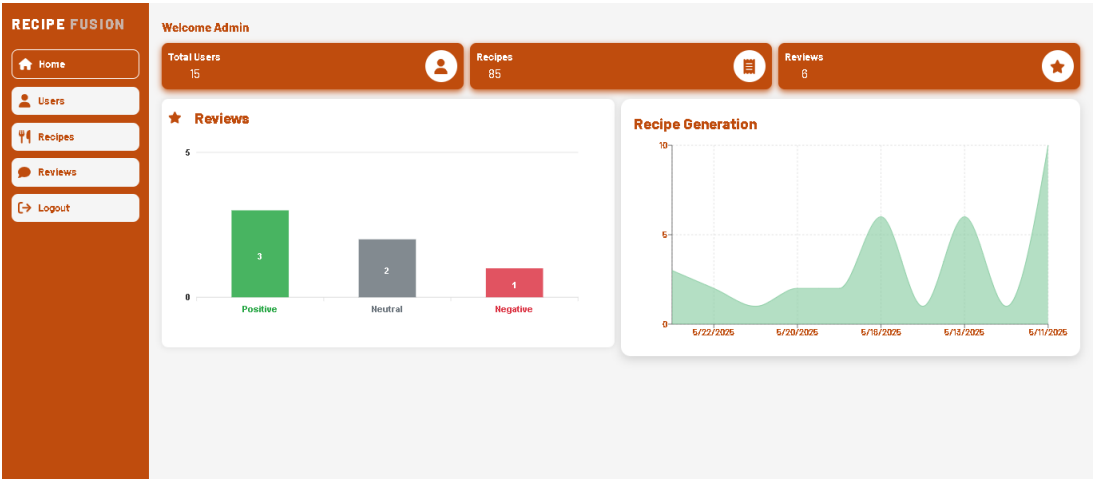


Fig 4.4.1 (d) – Home Screen

The Users Management Section displays a table of users with columns for Username, Email, and Actions. The Actions column includes View, Edit, and Delete buttons for each user.

Username	Email	Actions
Tayyaba Amjad	iamtayyabaamjad@gmail.com	View Edit Delete
Tayyaba Ejaz	tayyabaejaz123456@gmail.com	View Edit Delete
Ahsa	ahsan@gmail.com	View Edit Delete
Ikhshan	hlaiba83@gmail.com	View Edit Delete
user3	user3@gmail.com	View Edit Delete
userr	userr@gmail.com	View Edit Delete
admin	admin@gmail.com	View Edit Delete

Fig 4.4.1 (e) – Users Management Section

The Recipes Management Section displays a table of recipes with columns for Recipe Name, Image, Category, and Actions. The Actions column includes View and Delete buttons for each recipe.

Recipe Name	Image	Category	Actions
Red Lentil Soup with Chicken and Turnips		lunch	View Delete
Asparagus and Pea Soup: Real Convenience Food		antipasti	View Delete
Garlicky Kale		side dish	View Delete
Slow Cooker Beef Stew		lunch	View Delete
Red Kidney Bean Jambalaya		lunch	View Delete
Cauliflower, Brown			View Delete

Fig 4.4.1 (f) – Recipes Management Section

RECIPE FUSION

Home

Users

Recipes

Reviews

Logout

REVIEWS

Title	Review	Username	Email	Rating	Actions
Vegan Peach Cobbler Milkshake	Good	Mustafa	mustafa@gmail.com	neutral	Delete
Creamy zucchini and ham pasta	So tasty	tayyabaaa	tayyabaejz13375@gmail.com	positive	Delete
Coffee Cookies	Its a good recipe	tayyabaaa	tayyabaejz13375@gmail.com	neutral	Delete
Kc's Oreo Cheesecake	I like thi's recipe	userr	userr@gmail.com	positive	Delete
Couscous Biryani	good	M Ali	ali@gmail.com	positive	Delete
Banana Milo Milkshake	not good	Mustafa	mustafa@gmail.com	negative	Delete

Fig 4.4.1 (g) – Reviews Management Section

User Dashboard Screens:

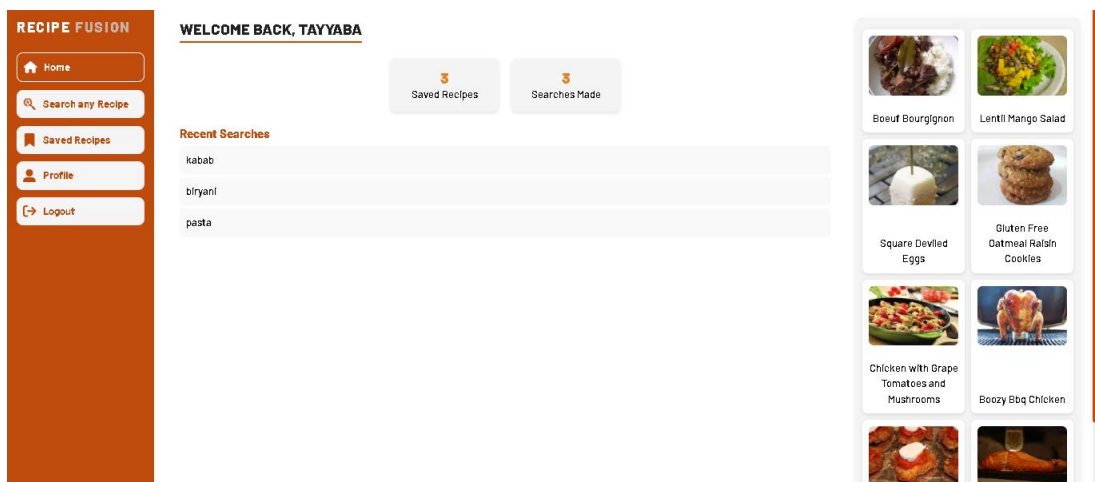


Fig 4.4.1 (h) – Home Screen

RECIPE FUSION Home Search any Recipe Saved Recipes Profile Logout	FIND YOUR FAVORITE RECIPES HERE Search by Name Search by Ingredients		
	SEARCH ANY RECIPE		
	Enter recipe name...	No Filter	No Cuisine
	Search		

Fig 4.4.1 (i) – Search Recipe Screen

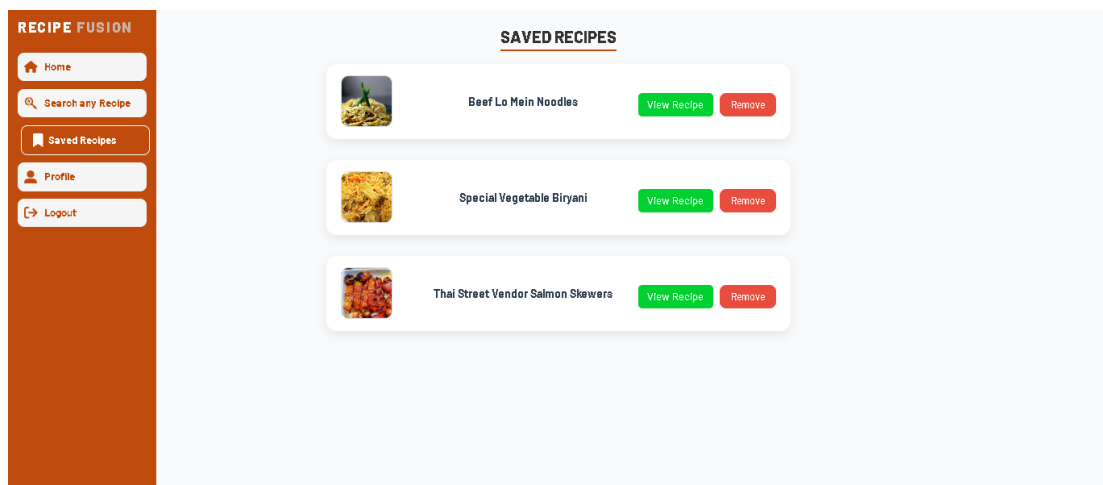


Fig 4.4.1 (j) – Saved Recipes

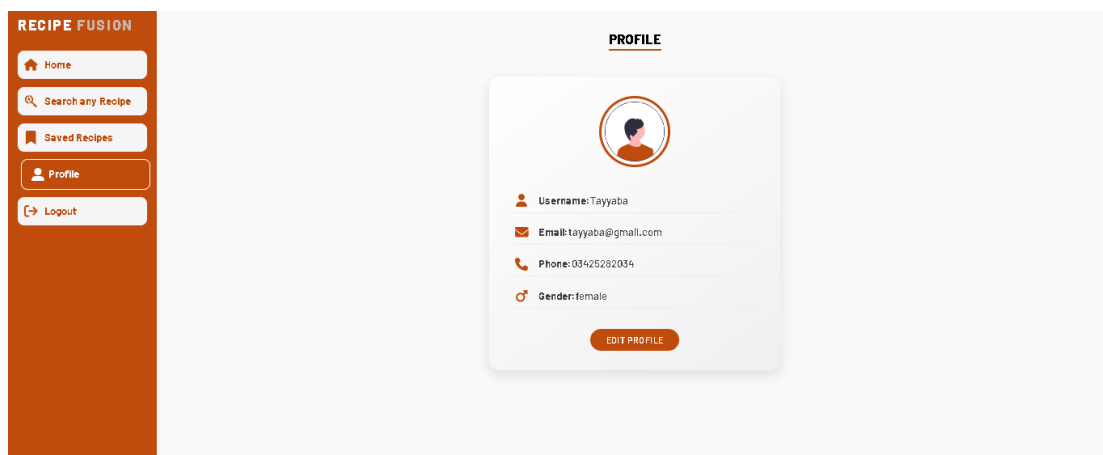


Fig 4.4.1 (k) – Profile Settings

4.5 View Model Architecture:

The View Model of Architecture outlines the structural framework of the RecipeFusion system, providing clarity for all stakeholders. This model breaks down the system into multiple views to explain behavior, data flow, and module interaction. It helps developers, users, and administrators understand how the system is structured and ensures it meets performance and usability standards. Feedback collected during modeling allowed for iterative improvements, ensuring the design remains user-focused and aligned with project requirements.

4.5.1 Logical View:

The logical view (as depicted in the corresponding class diagram) focuses on the core functional architecture of the RecipeFusion platform. It highlights the main components and their interactions that collectively support recipe recommendation, user personalization, and ingredient-based search functionality.

Key entities include:

User (General/Admin): Manages authentication, profile information, and role-based permissions. Users can search for recipes, save favorites, submit reviews, and manage dietary preferences.

Recipe: Represents complete meal ideas, including ingredients, preparation steps, cooking time, dietary tags (e.g., vegan, gluten-free), and user-contributed metadata like ratings and reviews.

Ingredient: Captures individual food items, allowing users to input available ingredients for matching recipes and enabling traceability features (e.g., source or origin, when applicable).

Review: Stores user feedback and ratings for recipes, contributing to the recommendation algorithm.

Saved Recipe: Allows users to bookmark or save recipes for future reference.

SearchHistory: Maintains records of a user's past searches to improve future suggestions and enhance personalization.

The logical view, represented through the class diagram, defines the structure and relationships among these core entities. It ensures coherent data flow and functional

consistency across the platform’s components. This view supports RecipeFusion’s operational goals by enabling personalized user experiences, efficient recipe recommendations, and scalable management of user and recipe data.

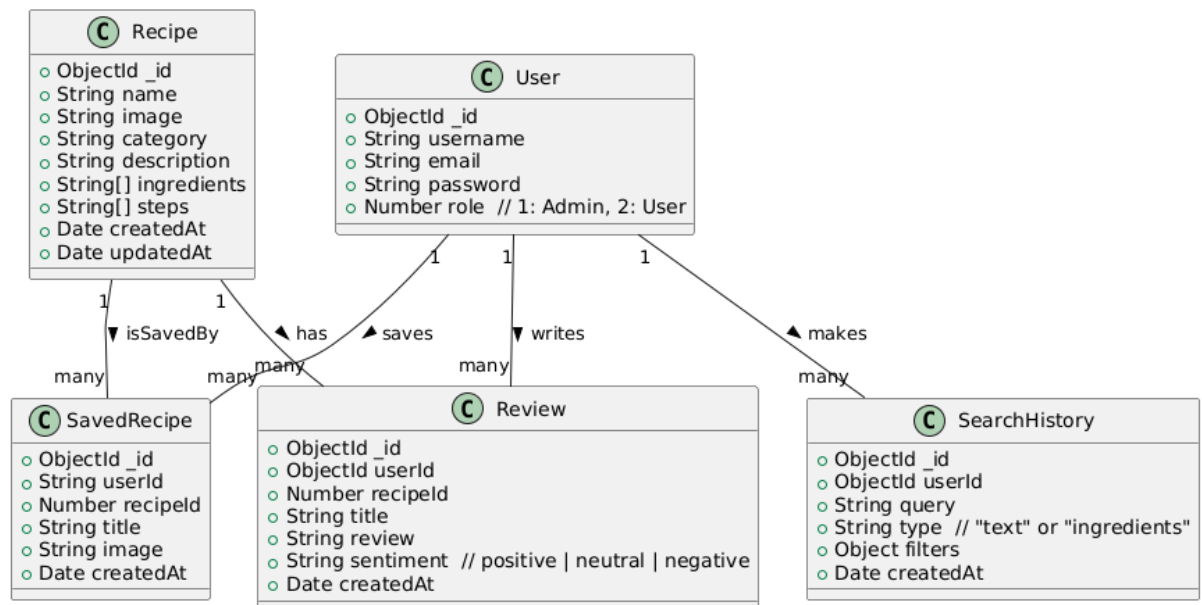


Fig 4.5.1 – Class Diagram

4.5.2 Process View:

The process view of the RecipeFusion platform outlines the key runtime behaviors, concurrent processes, and system interactions that facilitate personalized recipe discovery and user engagement. This view captures how the system components operate in real-time to ensure responsiveness, scalability, and efficiency.

Users begin their journey by registering or logging in through the frontend interface. Upon submission of login credentials, the backend (Node.js/Express) validates them, issues a JWT token for session management, and loads a role-based dashboard. Multiple users can authenticate and interact with the system concurrently without affecting each other’s sessions.

Once authenticated, users can input available ingredients, select dietary preferences, or explore recipes. The system concurrently queries the MongoDB database to retrieve matching recipe data, which is filtered based on criteria such as ingredients, cuisine type, or dietary restrictions. These search operations and filters are processed in parallel for seamless user experience.

Users may submit reviews, which are asynchronously stored and then aggregated to influence recipe ratings and recommendations. Simultaneously, search history is recorded to personalize future suggestions through collaborative filtering techniques.

Admins manage users, recipes, and reviews from a dedicated admin dashboard. They can perform operations like deleting inappropriate content, analyzing user search patterns, and managing the overall content repository. These actions occur independently from user-side interactions, maintaining system fluidity.

The media content (recipe images or videos) uploaded by users is processed and stored using Cloudinary, which handles file optimization and CDN-based delivery in the background, ensuring minimal delay during recipe browsing.

Smart search suggestions, caching mechanisms, and API request handling are designed to operate asynchronously, improving performance under high traffic. RecipeFusion also supports concurrency in actions like saving recipes, updating profiles, or browsing while interacting with other modules.

Throughout the system, the frontend (built with React.js) handles dynamic rendering and user interaction, while the backend manages API logic, authentication, and database communication. MongoDB ensures flexible data storage for recipes, users, and preferences, supporting concurrent read/write operations efficiently.

This process view illustrates RecipeFusion's responsive and event-driven architecture, where asynchronous operations and parallel processing ensure a seamless and personalized culinary experience for users across diverse roles and devices.

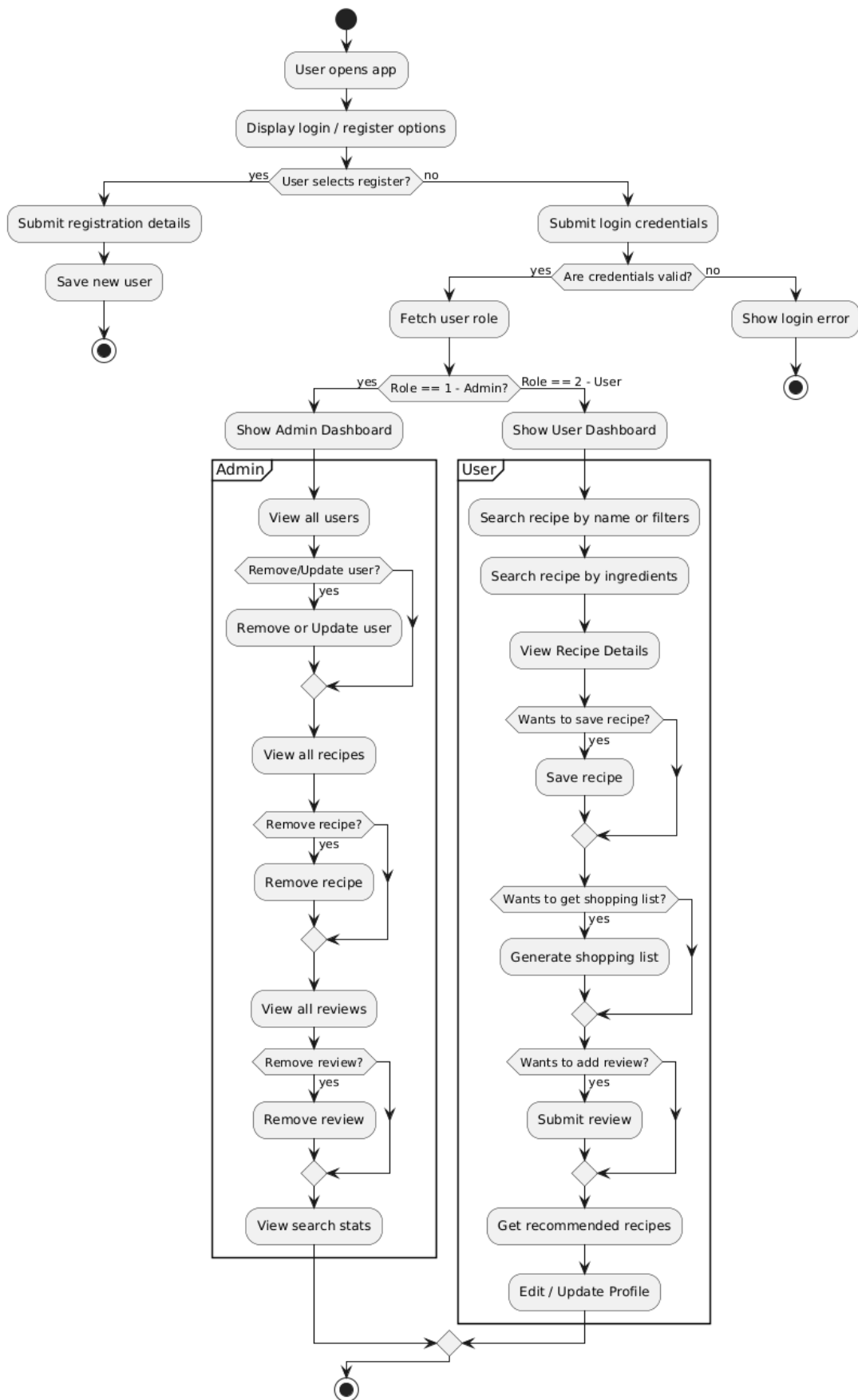


Fig 4.5.2 (a) – Activity Diagram

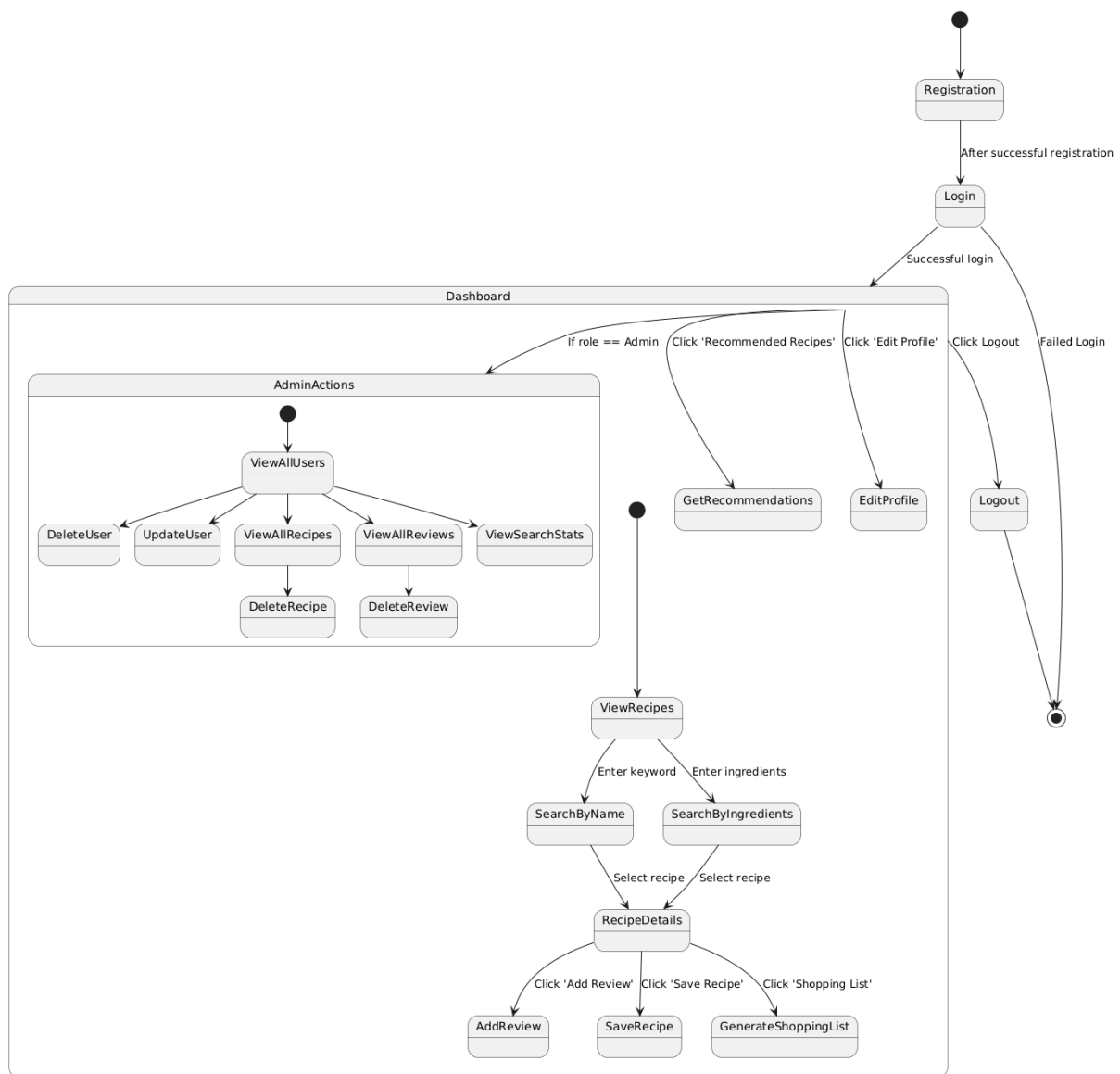


Fig 4.5.2 (b) – State Diagram

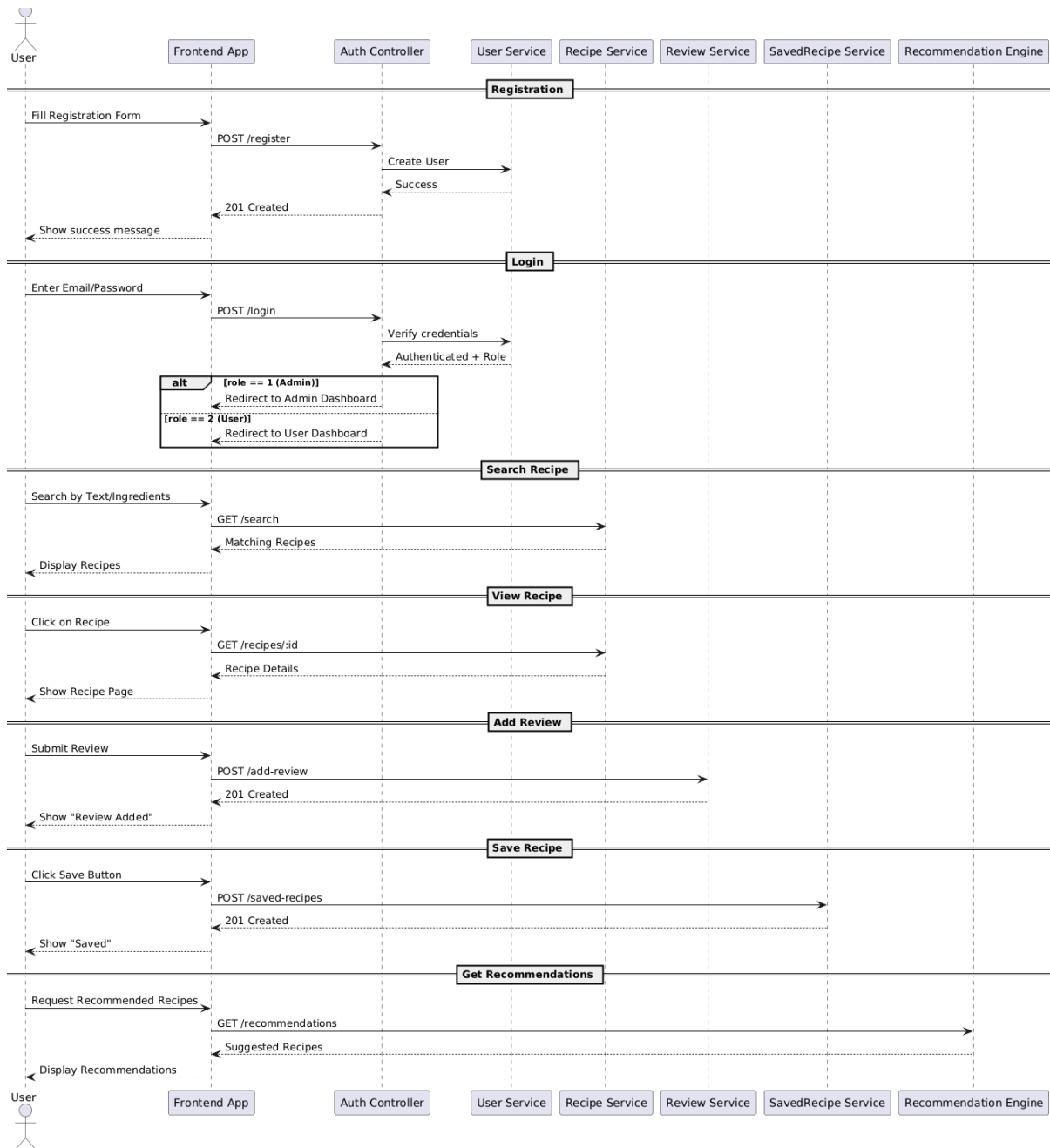


Fig 4.5.2 (c) – Sequence Diagram

4.5.3 Deployment View:

The Deployment View of the RecipeFusion platform illustrates how its core software components are distributed and hosted across different physical and cloud-based infrastructure layers to ensure high availability, performance, and scalability.

The project of the system, developed using React.js, and Node.js with express.js is deployed on a cloud hosting service such as Vercel providing responsive access to users across devices. This layer enables users to interact with features such as

ingredient-based recipe search, personalized recommendations, dietary filters, and recipe management dashboards.

A MongoDB database is hosted on a cloud database service like MongoDB Atlas, storing user information, recipe data, reviews, search history, and saved recipes. It ensures reliable, secure, and scalable data access with high availability and replication support.

The deployment infrastructure incorporates environmental configuration management, HTTPS-based secure communication, and role-based access control, ensuring data integrity and secure user interactions. Load balancing and auto-scaling are employed where necessary to maintain performance under varying traffic loads.

This deployment view captures the operational footprint of RecipeFusion across the cloud ecosystem, emphasizing modularity, scalability, and seamless user experience across browsers and devices.

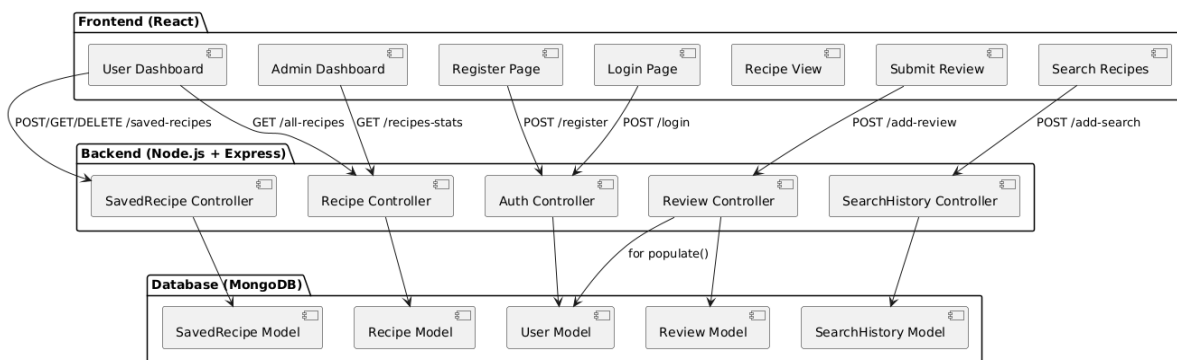


Fig 4.5.3 – Component Diagram

4.5.4 Physical View:

The Physical View of the RecipeFusion platform illustrates the underlying hardware, cloud infrastructure, and network components that support the operation of the system. RecipeFusion is deployed entirely in a cloud environment to ensure performance, scalability, and ease of access.

The frontend interface, developed using React.js, is hosted on cloud-based platforms such as Vercel which distribute static assets across global Content Delivery Networks (CDNs) to ensure fast and responsive user experiences on both desktop and mobile devices.

The backend services, built with Node.js and Express.js, are hosted on cloud infrastructure Vercel. These services manage API requests, business logic, authentication, and integration with external services such as cloud media storage and recommendation logic.

The MongoDB database is hosted via MongoDB Atlas, a fully managed cloud database service that offers high availability, backup, encryption, and automatic scaling. It stores key data including user profiles, recipes, ingredients, reviews, and search history.

All communication between clients and servers is secured using HTTPS protocols. Environment variables and secrets are securely managed using cloud platform configurations to protect sensitive credentials and API keys.

The physical infrastructure is designed with an emphasis on reliability, redundancy, and security, ensuring that the RecipeFusion platform remains available, scalable, and responsive for all users—from recipe seekers to administrators.

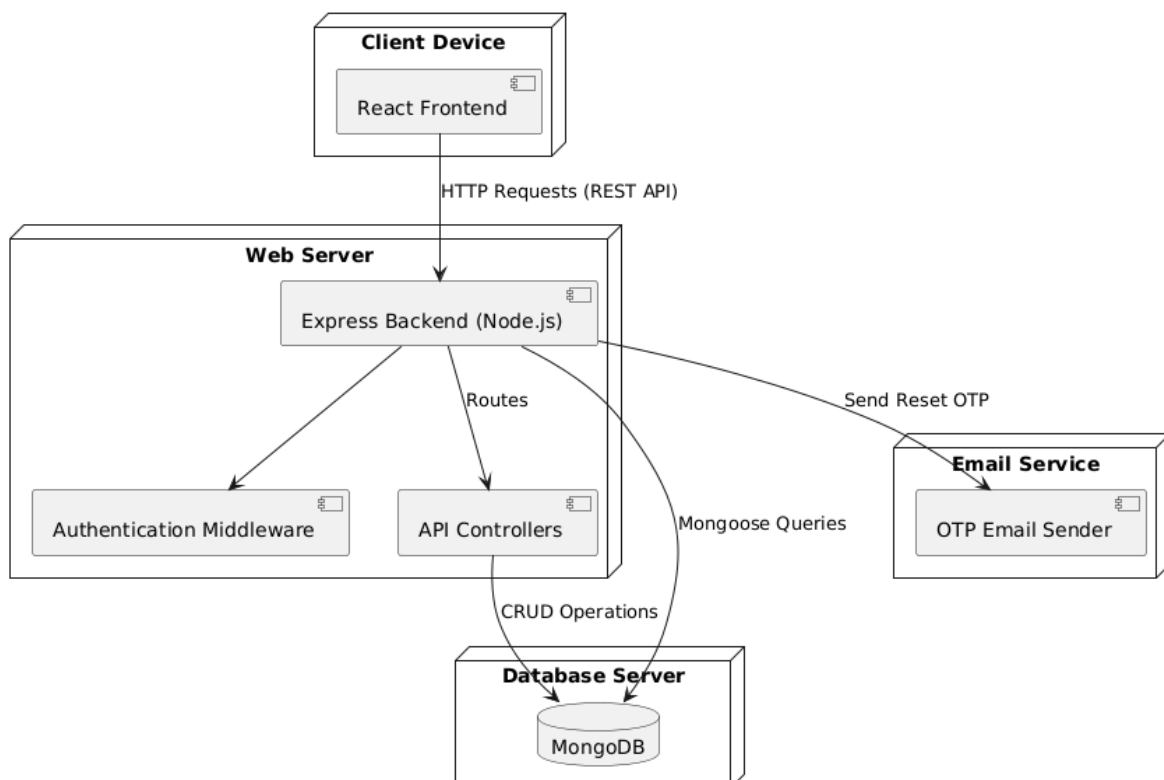


Fig 4.5.4 – Deployment Diagram

4.6 Entity Relationship Diagram (ERD):

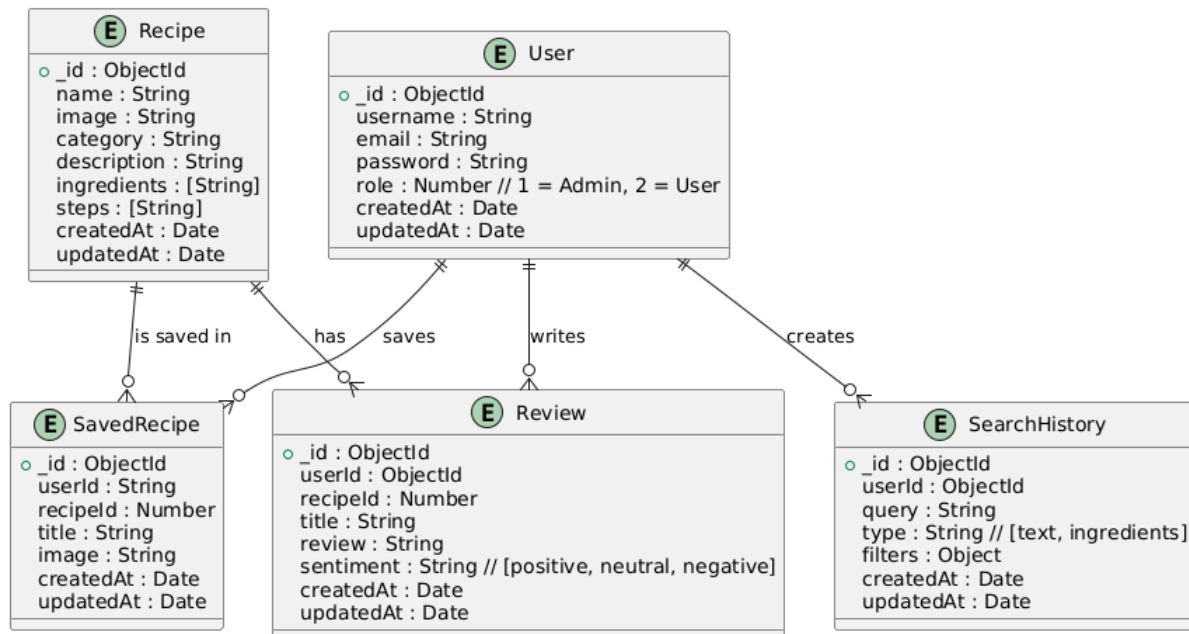


Fig 4.6 – ERD Diagram

4.7 Summary

In this chapter, we presented the system modeling and design of the RecipeFusion platform, emphasizing its modular architecture and user-centric approach powered by modern web technologies. The chapter began with an introduction to system modeling and a description of the bottom-up design approach, which guided the incremental development of key modules such as user management, recipe handling, ingredient-based search, and personalized recommendations.

The system design follows the 4+1 View Model of Architecture, which provides a comprehensive representation of the platform through its logical, process, development, and physical views, each supplemented by UML diagrams like class, activity, sequence, component, and deployment diagrams.

These models illustrate the flow and interaction between components such as user registration and login, ingredient-based recipe search, filtering based on dietary preferences, review submission, and admin management of platform content. The architecture also supports advanced features like nutritional data handling, review analysis, and user behavior tracking.

The chapter concludes with the deployment and physical infrastructure of the system, demonstrating how cloud-based services like Vercel, MongoDB Atlas, ensure scalability, performance, and secure access for both end users and administrators. This architecture supports the platform's vision of promoting healthy eating habits and personalized culinary experiences through a seamless and intelligent digital solution.

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